

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular & Supplementary Winter Examination-2023 Course: B. Tech. Branch: Mechanical & Allied Engineering Semester: III Subject Code & Name: BTMC302 / Fluid Mechanics Max Marks: 60 Date:04-01-24 Duration: 3Hr.			
<i>Instructions to the Students:</i> 1. <i>All the questions are compulsory.</i> 2. <i>The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.</i> 3. <i>Use of non-programmable scientific calculators is allowed.</i> 4. <i>Assume suitable data wherever necessary and mention it clearly.</i>			
		(Level/CO)	Mark s
Q. 1	Solve Any Two of the following.		(12)
A)	Describe any three terms: Pascal's Law, Hydrostatic Law, Total pressure, Center of pressure	(Understand / CO1)	(6)
B)	A plate 0.125 mm distant from a fixed plate, moves at 60 cm/s and requires a force of 2 N per unit area i.e., 2 N/m² to maintain this speed. Determine the fluid viscosity between the plates in poise.	(Apply / CO1)	(6)
C)	Explain the phenomenon of capillarity. Obtain the expression for capillary rise of a liquid in circular glass tube of diameter 'd'.	(Understand / CO1)	(6)
Q.2	Solve Any Two of the following.		(12)
A)	Derive the Bernoulli's equation with the help of Euler's equation. Also discuss the assumptions made in Bernoulli's equation.	(Apply / CO2)	(6)
B)	Define following types of flows (Any three). Steady and unsteady flow Uniform and non-uniform flow Laminar and Turbulent flow Compressible and incompressible flow	(Understand / CO2)	(6)

C)	An orifice meter with orifice diameter 10 cm is inserted in a pipe of 20 cm diameter. The pressure gauges fitted upstream and downstream of the orifice meter gives readings of 19.62 N/cm ² and 9.81 N/cm ² respectively. Co-efficient of discharge for the orifice meter is given as 0.6. Find the discharge of water through pipe.	(Apply / CO2)	(6)
Q. 3	Solve Any Two of the following.		(12)
A)	A crude oil of viscosity 0.97 poise and relative density 0.9 is flowing through a horizontal circular pipe of diameter 100 mm and of length 10 m. Calculate the difference of pressure at the two ends of the pipe, if 100 kg of the oil is collected in a tank in 30 seconds.	(Apply / CO3)	(6)
B)	Explain major energy loss and minor energy losses in pipes.	(Understand / CO3)	(6)
C)	Define and explain the following terms with the help of suitable sketch. Hydraulic gradient line Total energy line	(Understand / CO3)	(6)
Q.4	Solve Any Two of the following.		(12)
A)	Differentiate between Stream line body and bluff body Friction drag and pressure drag	(Understand / CO4)	(6)
B)	Describe the following term with the help of neat sketch. Laminar boundary layer, turbulent boundary layer, laminar sublayer and boundary layer thickness.	(Understand / CO4)	(6)
C)	What is drag and lift? Explain types of drags.	(Understand / CO4)	(6)
Q. 5	Solve Any Two of the following.		(12)
A)	The resisting force R of a supersonic plane during the flight can be considered as dependent upon the length of the aircraft l , velocity v , air viscosity μ , air density ρ and bulk modulus of air K . Express the functional relationship between these variables and the resulting force.	(Apply / CO5)	(6)

B)	What is dimensionless number? Define any four dimensionless numbers.	(Understand / CO5)	(6)
C)	What is repeating variable? How are the repeating variables selected for dimensional analysis?	(Understand / CO5)	(6)
	*** End ***		

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